In [1]:

```python
# Q1. E-Commerce Order Analysis

# Given order data
orders = [
    {
        'id': 101,
        'customer': 'Asha',
        'status': 'DELIVERED',
        'items': [('keyboard', 2, 1499.0), ('mouse', 1, 599.0)]
    },
    {
        'id': 102,
        'customer': 'Ravi',
        'status': 'CANCELLED',
        'items': [('monitor', 1, 10999.0)]
    },
    {
        'id': 103,
        'customer': 'Asha',
        'status': 'DELIVERED',
        'items': [('cable', 5, 199.0)]
    },
    {
        'id': 104,
        'customer': 'Meera',
        'status': 'SHIPPED',
        'items': [('keyboard', 1, 1499.0), ('monitor', 1, 10999.0)]
    },
    {
        'id': 105,
        'customer': 'Ravi',
        'status': 'DELIVERED',
        'items': [('mouse', 3, 599.0)]
    }
]


# Function to analyse orders
def analyse(orders):

    # Create empty dictionaries
    revenue = {}
    products = {}
    customer_status = {}

    # Read each order
    for order in orders:

        # Get customer name and order status
        customer = order['customer']
        status = order['status']

        # Check customer delivery status
        if customer not in customer_status:
            customer_status[customer] = True

        if status != 'DELIVERED':
            customer_status[customer] = False
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        # Ignore cancelled orders
        if status == 'CANCELLED':
            continue

        # Read products from each order
        for item in order['items']:

            # Get product details
            product = item[0]
            quantity = item[1]
            price = item[2]

            # Calculate total amount
            amount = quantity * price

            # Calculate customer revenue
            if customer not in revenue:
                revenue[customer] = 0

            revenue[customer] = revenue[customer] + amount

            # Calculate product quantity
            if product not in products:
                products[product] = 0

            products[product] = products[product] + quantity

    # Find top 2 selling products
    top_products = sorted(
        products.items(),
        key=lambda x: (-x[1], x[0])
    )[:2]

    # Find fully delivered customers
    fully_delivered = []

    for customer in customer_status:

        if customer_status[customer] == True:
            fully_delivered.append(customer)

    # Sort customer names
    fully_delivered.sort()

    # Store all results in dictionary
    result = {
        'revenue': revenue,
        'top_products': top_products,
        'fully_delivered': fully_delivered
    }

    # Return final result
    return result


# Call the function and print answer
answer = analyse(orders)
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print("Q1 Answer:")
print(answer)
```

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Q1 Answer:
{'revenue': {'Asha': 4592.0, 'Meera': 12498.0, 'Ravi': 1797.0}, 'top_products':
[('cable', 5), ('mouse', 4)], 'fully_delivered': ['Asha']}
```

In [2]:

```
# Q2. Customer Data Cleaning

# Given customer records
lines = [
    ' asha VERMA ; ASHA.V@Example.COM ; +91 98765-43210 ;pune ',
    'ravi kumar;ravi@@corp.in;9123456780;Mumbai',
    'meera;meera@mail;9123456780;Delhi',
    'john doe; john@x.org ;12345; Goa',
    'Neha Rao;neha.rao@firm.co.in;919876501234;nashik',
    'amit;amit@site.com;9876543210'
]


# Function to clean customer data
def clean_customer(raw):

    # Split record using semicolon
    data = raw.split(';')

    # Check whether there are exactly 4 fields
    if len(data) != 4:
        return None

    # Remove extra spaces
    name = data[0].strip()
    email = data[1].strip().lower()
    phone = data[2].strip()
    city = data[3].strip()

    # Convert name into title case
    name = name.title()

    # Split name into words
    words = name.split()

    # Check name is not empty
    if len(words) == 0:
        return None

    # Check that name contains only letters
    for word in words:

        for ch in word:

            if not ch.isalpha():
                return None

    # Check city is not empty
    if city == '':
        return None

    # Convert city into title case
    city = city.title()
```

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    # Check email contains exactly one @
    if email.count('@') != 1:
        return None

    # Split email into username and domain
    parts = email.split('@')

    username = parts[0]
    domain = parts[1]

    # Check username is not empty
    if username == '':
        return None

    # Check domain contains a dot
    if '.' not in domain:
        return None

    # Check dot is not at starting or ending position
    if domain[0] == '.' or domain[-1] == '.':
        return None

    # Remove spaces and hyphens from phone
    phone = phone.replace(' ', '')
    phone = phone.replace('-', '')

    # Remove +91 from phone number
    if phone.startswith('+91'):
        phone = phone[3:]

    # Remove 91 if phone number has 12 digits
    elif len(phone) == 12 and phone.startswith('91'):
        phone = phone[2:]

    # Check phone number has exactly 10 digits
    if len(phone) != 10:
        return None

    # Check all characters are digits
    for digit in phone:

        if not digit.isdigit():
            return None

    # Check phone starts with 6, 7, 8 or 9
    if phone[0] not in '6789':
        return None

    # Create dictionary of valid customer
    customer = {
        'name': name,
        'email': email,
        'phone': phone,
        'city': city
    }

    # Return cleaned customer data
    return customer
```

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# Count invalid records
rejected = 0

# Read each customer record
for record in lines:

    # Call function
    result = clean_customer(record)

    # Print valid customer records
    if result is not None:
        print(result)

    # Count invalid records
    else:
        rejected = rejected + 1

# Print total rejected records
print("Rejected:", rejected)
```

```
{'name': 'Asha Verma', 'email': 'asha.v@example.com', 'phone': '9876543210', 'cit
y': 'Pune'}
{'name': 'Neha Rao', 'email': 'neha.rao@firm.co.in', 'phone': '9876501234', 'cit
y': 'Nashik'}
Rejected: 4
```

In [ ]: